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Original Research

A Longitudinal Analysis of Boundary-Spanning Leadership and Absorptive Capacity as Drivers of Job Embeddedness

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Abstract: The purpose of this research is to explore a novel model whereby boundary-spanning leadership (BSL) enhances nurses' absorptive capacity (ACAP), ultimately leading to increased job embeddedness (JE) during the COVID-19 pandemic. An autoregressive cross-lagged model was employed to examine the causal relationships among BSL, ACAP, and JE. Data were collected through three waves of surveys conducted at three-month intervals, with a final sample of 297 nurses in Chinese hospitals. This longitudinal design enabled the investigation of both the direct and indirect effects of BSL on JE, mediated by ACAP, as well as the bidirectional influence between BSL and ACAP over time. First, the autoregressive effects confirmed the temporal stability of BSL, ACAP, and JE, indicating that these constructs remained consistently valid in real-world settings over time. Second, the cross-lagged analysis validated that BSL, by fostering ACAP, indirectly enhanced JE, while BSL had no direct effect on JE. Finally, the bidirectional relationship between BSL and ACAP revealed a reciprocal dynamic between these two constructs. This research introduces a unique model that integrates BSL, ACAP, and JE, addressing gaps in prior literature and providing meaningful insights through rigorous analysis using longitudinal data and an autoregressive cross-lagged model. We suggested that hospital administrators and HR leaders foster ACAP through regular feedback sessions and workshops that promote knowledge sharing and the exploration of new ideas, and further emphasized the role of BSL within tailored training programs.

Keywords: *Boundary-spanning Leadership, Absorptive Capacity, Job Embeddedness, Autoregressive Cross-lagged Model*

Introduction

The COVID-19 pandemic has intensified the dynamism and complexity of the business environment as the previously unexperienced phenomena have become established as part of a new normal. Amidst increasing complexity and competition, firms endeavor to overcome these challenges through transforming their organizational systems (Benoliel and Somech 2014). The endeavors involve organizational responses to environmental changes such as enhancing task efficiency through promoting cross-functional collaboration and communication, networking with external stakeholders, facilitating knowledge transfer, and engaging in community outreach. These diverse attempts within firms at both group and individual levels can be interpreted as boundary-spanning behaviors (hereafter referred to as

BSB). The interest regarding organizational external activities has increased following the introduction of the BSB concept by social psychologists (Leicht-Deobald et al. 2025; Somech and Khalaili 2014). BSB helps connect and coordinate with the external environment while mitigating the shocks and uncertainties (Ancona and Caldwell 1992; Zhang and Li 2020). Consequently, the success of BSB leads to performance in seizing and exploiting knowledge and information from within and outside the organization boundary.

Modern leaders frequently face increasing pressure to fulfill their responsibilities with heightened urgency. BSB has long been recognized as a role that managers should perform (e.g., Leicht-Deobald et al. 2022; Prysor and Henley 2017). Managers tasked with engaging external stakeholders are expected to excel in negotiations and information exchange (Mell et al. 2022). Moreover, managers with the authority or responsibility to interact with external stakeholders are likely to be proficient in negotiations and information exchange (Takanashi and Lee 2018). In this regard, the concept of boundary-spanning leadership (hereinafter, BSL) has recently received attention, which encompasses the role of managers in supporting and facilitating employees' BSB and is essential for effective boundary management (Mitchell and Nicholas 2006; Hans et al. 2023). Boundary-spanners with BSL, who possess sufficient knowledge and experience to understand and communicate contextual information in both the internal and external domains of the boundary, are critical for effectively adapting to environmental changes (Setiadi and Widodo 2024; Xue and Woo 2022). Therefore, the pandemic would also provide an opportune context to elucidate the impacts of BSL.

However, studies examining the impact of BSL on adapting to the pandemic are scarce (Setiadi and Widodo 2024). To address this research gap, this study aimed to explore the experiences of hospital nurses who encountered confusion and excessive workload during the pandemic as well as to verify the impact and mechanisms of BSL on subordinates' performance. We investigated the impact of BSL on nurses' absorptive capacity (hereinafter, ACAP), which encompasses acquiring, sharing, and utilizing knowledge and information both internally and externally, and its impact on job embeddedness (hereinafter, JE). In the case of ACAP, research at the individual level is still in its early stages, indicating a lack of extensive studies (Yildiz et al. 2019; Tian and Soo 2018). Currently, there is no research linking nurses' BSB in the hospital context to ACAP. While there are studies indicating that JE—defined as the degree to which individuals are connected to and remain with their job and organization—is influenced by leadership styles such as servant leadership (Fan et al. 2024; Faraz et al. 2023) and ethical leadership (Nguyen et al. 2023), there remains a lack of research examining its relationship with BSL. The findings of this investigation are expected to offer valuable implications regarding the roles of BSL as a novel leadership approach to successfully adapt to environmental shifts such as those experienced during the pandemic.

Theoretical Review and Hypotheses

Boundary-Spanning Leadership

The concept of "boundary-spanning" focuses on the importance of absorbing, sharing, and transmitting information and knowledge across organizational boundaries to foster innovation and performance (Tushman and Scanlan 1981a, 1981b; Marrone et al. 2007). Organizations in static environment tend to rely on well-established rules, which may result in limited engagement of BSB (Zou and Ingram 2013). In contrast, when the environment surrounding the organization is uncertain and rapidly changing, BSB is likely to increase to mitigate the impact of such changes (Zhao and Anand 2013). For innovative organizations that need to exchange information and interact with the external environment, managers' support is crucial for the successful implementation of BSB. This is because managers with BSL play an encouraging role in establishing relationships with external organizations, providing a platform for exchanging knowledge, information, and ideas to foster commitment across boundaries (Salem et al. 2017; Mell et al. 2022). Researchers who consider boundary spanning as a critical element of successful leadership have conceptualized BSL as the ability to generate alignment and commitment across organizational boundaries (Mell et al. 2022; Benoliel and Somech 2014). In other words, BSL refers to a manager's actions in pursuing alignment and commitment across the boundaries of an organization to achieve higher objectives (Ernst and Chrobot-Mason 2011; Cross et al. 2015).

Although academic consensus on the definition of BSL is lacking, the framework proposed by Cross et al. (2013) is the most widely accepted, which subdivides BSL into six stages (see Figure 1). First, buffering entails activities that protect the team from threats and changes in the external environment and solidify team identity. Second, reflecting encompasses activities that comprehend and acknowledge the identities of other groups outside the boundary. Buffering forms the foundation for stable boundary activities, while reflecting builds cooperation with other groups based on mutual respect. Third, connecting involves activities that establish relationships that transcend existing boundaries, leading to the discovery of new shared values such as mutual benefit and common goals. Fourth, mobilizing consists of activities that discard existing boundaries and identities to restructure more productive new boundaries to achieve new shared values. Fifth, weaving, akin to weaving fabric with warp and weft threads, comprises activities that recognize the unique roles of other teams and integrate their capabilities. Lastly, transforming encompasses activities aimed at rediscovering innovative possibilities and alternatives emerging from newly consolidated boundaries.

These six elements are subsequently categorized into three strategies, each containing a pair of activities: managing boundaries, forging common ground, and discovering new frontiers (see Figure 1). The first strategy, managing boundaries, involves reflecting and buffering mechanisms aimed at fostering safety and respect across boundaries. The second strategy, forging common ground, is about mobilizing and connecting in order to cultivate trust and engagement across boundaries. The final strategy, discovering new frontiers, nurtures the creation of a "team of teams" through transforming and weaving, thereby enabling breakthrough innovation and strategic reinvention.

By effectively managing inter-organizational boundaries, BSL facilitates the flow of information, resources, and perceptions among individuals participating in the innovation process (Fleming and Waguespack 2007). Boundaries inherently exhibit an ambivalent nature, characterized by the simultaneous existence of cooperative and competitive tensions. In such contexts where judgment and coordination are challenging, BSL provides direction to organizational members and facilitates their engagement (Cross et al. 2015). Furthermore, BSL encourages subordinates to participate in innovation, thereby contributing to the smooth flow of information and resources between groups (Chreim et al. 2013; Nazarzadeh Zare and Ghoraishi Khorasgani 2022). Consequently, BSL is expected to have a positive impact on intergroup cooperation, collaboration, learning among subordinates (Xue and Woo 2022; Salem et al. 2017), and their absorptive capacity (ACAP). However, research on the relationship between BSL and subordinates' ACAP remains limited.

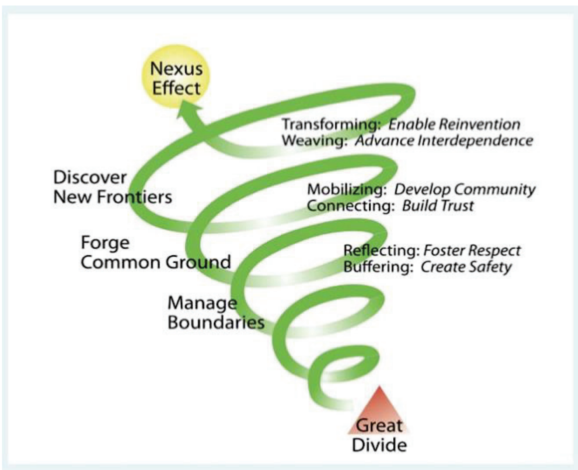

Figure 1: Boundary-Spanning Activities That Collectively Serve as a Powerful Catalyst for Action
Source: Ernst and Chrobot-Mason 2010

Absorptive Capacity

Although ACAP was initially conceptualized at the organizational level, recent studies (e.g., Yildiz et al. 2019; Yao and Chang 2017) have increasingly explored the concept at the individual level. ACAP originated from Schumpeterian economic theory, which emphasizes the impact of a firm's R&D capabilities on economic performance (Zahra and George 2002) and was subsequently developed in conjunction with cognitive psychology and organizational learning theories (Yao and Chang 2017). Cohen and Levinthal (1990) conceptualized and diffused ACAP as the ability to acquire, comprehend, and apply diverse information and knowledge from external sources to achieve performance. However, focusing excessively on the organizational level has been criticized for obscuring the role of individuals in developing ACAP (Yildiz et al. 2019). This lack of attention to individual-level ACAP has led to the misconception that ACAP exists only at the organizational level, overlooking the role of individuals in developing and maintaining ACAP (Lane et al. 2006; Tian and Soo 2018).

Consequently, ACAP has expanded beyond corporate R&D activities to encompass employees' attributes such as knowledge depth and breadth, prior learning experiences, and problem-solving abilities (Tsai 2022). Individuals' exploration and exploitation of new knowledge can induce changes in ways of thinking and behaviors that contribute to productivity (Zahra and George 2002). ACAP at the individual level positively influences personal innovation behavior through employee agility (Da Silva and Davis 2011). Dolmark et al. (2022) found that individual ACAP has a positive effect on learning behavior in a technological context. Furthermore, variations in ACAP among individuals affect personal outcomes, such as learning, innovation, and productivity (Schweisfurth and Raasch 2018; Yildiz et al. 2019). This implies that ACAP is a critical factor related to various BSB at the individual level.

Individual ACAP refers to the ability to recognize, assimilate, and exploit diverse information and knowledge from external sources beyond boundaries (Yildiz et al. 2019). Recognition entails individuals actively seeking new external knowledge and acknowledging its associated benefits. Assimilation measures how effectively individuals integrate new knowledge with their established understanding. Exploitation evaluates the extent to which the assimilated knowledge is utilized for commercial purposes. The attributes of individual ACAP positively influence various outcomes such as learning, innovation, and commitment. Furthermore, this study specifically proposes that employees' ACAP stimulates cognitive changes that enhance their job embeddedness (JE).

Job Embeddedness

Employees who proactively cultivate ACAP demonstrate enhanced adaptability within their organization by aligning themselves with the organization's values (Pratoom 2022). Furthermore, these employees prioritize the development of robust networks both internally and externally, thereby becoming fully embedded in their roles. JE, a concept rooted in organizational psychology, reflects the degree to which an employee is interconnected and engaged in their job, organization, and work environment (Mitchell et al. 2001). JE refers to the extent of an employee's engagement within their work context, accounting for factors such as job satisfaction, collegial relationships, and opportunities for growth (Crossley et al. 2007).

The concept of JE emerged in response to the limitations of existing research on employee turnover (Ng and Feldman 2007). Higher levels of JE are associated with reduced turnover rates, increased organizational commitment, and enhanced job satisfaction and performance (Tanova and Holtom 2008). Notably, a key characteristic of JE is the exclusion of emotional aspects from its measurement, distinguishing it from the concept of organizational commitment, which encompasses emotional responses such as strong attachment to or identification with the organization (Kiazad et al. 2015). JE encapsulates the cognitive dimensions of an employee's abilities and fit, whereas organizational commitment addresses the emotional facets (Zhou et al. 2021).

JE comprises three fundamental components in organizational research: links, fits, and sacrifices (Mitchell et al. 2001). First, links denote the formal and informal connections an employee establishes with other individuals or teams within the organization and the wider communities beyond the organization's boundaries. Intra-organizational links manifest as interdependencies between an individual's job outcomes and those of their colleagues, or through collaborative relationships with coworkers across various departments while undertaking multiple roles. Extra-organizational links emerge through personal pursuits such as hobbies or religious activities, facilitating a deeper connection to the community and reinforcing the individual's sense of belonging. Second, fits represent the degree of alignment between an employee's qualifications or interests and job demands necessary for performance of duties. The former includes an individual's skills, knowledge, values, and career aspirations, while the latter can be exemplified by job requirements and the organization's culture and mission. Third, sacrifice encompasses the perceived tangible and intangible costs that an employee incurs upon leaving their current job or organization. These costs include the loss of career development opportunities, amicable relationships, job stability, and more.

While most research on JE is fixated on its association with turnover, studies on the antecedents of JE remain underdeveloped. Expanding upon the limited research on the antecedents of JE, we investigated the effects of BSL and ACAP.

Hypothetical Framework

This study aims to investigate the interconnected causal associations among BSL, ACAP, and JE by formulating a set of research hypotheses (see Figure 2). Based on social learning theory (Bandura 1977), managers serve as role models for their subordinates, demonstrating how expected behaviors can be achieved through intellectual stimulation (Naqshbandi and Tabche 2018). The theory is particularly relevant to this study because it emphasizes that subordinates learn behaviors and attitudes not only through direct instruction but also by observing and modeling the behaviors of influential figures such as managers. Leadership plays a crucial role in shaping creativity and innovation, which are closely associated with ACAP (Mumford et al. 2003). BSL, as a form of leadership, encourages subordinates to explore new knowledge and fosters knowledge sharing within the organization (Salem et al. 2017; Shah et al. 2018), facilitating a smooth flow of information and resources across organizations (Chreim et al. 2013). Therefore, it can be inferred that BSL can positively impact subordinates' ACAP.

ACAP activated by BSL requires connections and collaboration within and outside the organization. Subordinates' ACAP refers to the ability to acquire and assimilate external knowledge and transform it into work processes (Tsai 2022; Enkel et al. 2017). Therefore, ACAP shares many aspects with BSB, as both involve subordinates gathering and utilizing information to learn and improve their work. ACAP encompasses shared feedback and discussions about problems and errors (Tian and Soo 2018; Dolmark et al. 2021). This collaborative process can enhance subordinates' cognitive interdependence and their collective understanding of environmental changes. When subordinates recognize the importance of sharing knowledge and information, such recognition builds a psychological contract (Robinson 1996). Such informal contracts can influence individual engagement, commitment, loyalty, and turnover (Herrera and De Las Heras-Rosas 2021; Sheehan et al. 2019), thereby enhancing JE.

Hypothesis 1: ACAP will mediate the relationship between BSL and JE.

Leadership plays a crucial role in shaping employees' attitudes and commitment. In particular, effective leadership helps reduce turnover by fostering a supportive and engaging work environment. Transformational leadership and BSL are closely aligned in that both aim to inspire and empower subordinates to pursue goals beyond their immediate tasks. While transformational leadership focuses broadly on motivating followers through vision, intellectual stimulation, and individualized consideration, BSL specifically highlights the importance of supporting boundary-spanning behaviors that facilitate knowledge sharing and collaboration across organizational lines. Thus, BSL can be viewed as a context-specific extension of transformational leadership, particularly relevant in environments that demand external engagement and adaptive learning. According to transformational leadership theory (Yammarino and Dubinsky 1994), BSL offers subordinates a vision of BSB, encourages participation in decision-making, and fosters trust, loyalty, and satisfaction in problem-solving situations (Harris et al. 2011; Khalid et al. 2021). Such a nurturing environment helps develop mutually beneficial, long-term relationships between managers and subordinates, going beyond formal work interactions to create deeper partnerships (Caramanica and O'Rourke 2021). BSB, when accompanied by support, communication, and emotional engagement from leaders, is related to JE (Harris et al. 2011). Thus, we present the following hypothesis based on these arguments:

Hypothesis 2: BSL will have a positive impact on job embeddedness.

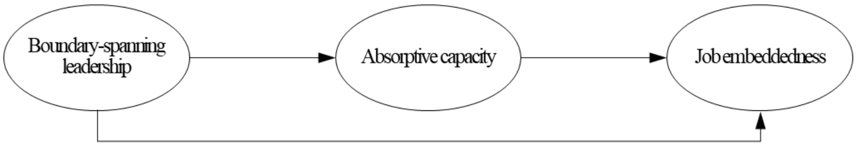

Figure 2. Hypothetical Model

Method and Materials

This research seeks to explore the experiences of hospital nurses who faced confusion and heightened workloads amidst the unprecedented conditions of the COVID-19 pandemic. Adhering to research ethics, it was carried out through self-report questionnaires, involving participants from eleven hospitals in China. To assess the difficulties arising from the pandemic and the changes in job roles, the survey deliberately excluded participants who had been employed for five years or longer.

Our analytical strategy employs an autoregressive cross-lagged model to obtain more accurate results (Ployhart and Vandenberg 2010; Schlueter et al. 2007). Cross-sectional research may have limitations in evaluating genuine causality effectively. Temporal order, one of the conditions of causation, refers to the requirement that the cause must precede the effect in time. The reason temporal order cannot be established in cross-sectional studies is that data is collected at a single point in time. In order to gather longitudinal data, we followed the recommendation of Singer and Willett (2003) to collect data at least three times to observe systematic changes in all variables over time. The panel consisted of the same participants who underwent repeated measurements at three different time points, spaced at three-month intervals, from January 2022 to February 2023.

Initially, 425 individuals (100%) participated in the survey. Subsequently, the participation rate decreased to 329 (75.6%) at Time 2 and further declined to 297 (68.3%) at Time 3. These numbers represent the participants remaining after excluding those who did not complete certain items. After careful consideration of the research model and analysis methods used in this study, we conclude that the final sample size of 297 participants is adequate. All survey measures and protocols adhered to the ethical standards set forth by the American Psychological Association. Written informed consent was obtained from all participants prior to data collection. Formal ethical approval was waived in accordance with the exemption regulations of the Korean Ethics Committee, as the physical and psychological risks to participants were deemed minimal and the data collected did not contain any personal identifiers.

The main factors were assessed using the measurement instrument on a 6-point scale ranging from 1 (“strongly disagree”) to 6 (“strongly agree”). Respondents expressed their level of agreement with each statement using a Likert scale. BSL was measured using twenty-four items developed by Palus et al. (2014), which assess managers’ abilities to navigate across six dimensions: buffering, reflecting, connecting, mobilizing, weaving, and transforming. Each dimension is assessed by four items, totaling twenty-four items on the scale. ACAP refers to an individual's ability to recognize, assimilate, and utilize new information and knowledge across boundaries. The scale created by Yildiz et al. (2019) consists of fourteen items divided into three sub-dimensions: recognition (four items), assimilation (six items), and exploitation (four items). JE is defined as the situational and cognitive factors that keep individuals connected to people or work-related issues (Crossley et al. 2007). The scale for JE, developed by Mitchell et al. (2001), includes fourteen items: six items related to fit, four items to sacrifices, and four items to links.

To reduce the complexity of the research model caused by an excessive number of measurement items, which could potentially diminish model fit (Bandalos 2002; Little et al. 2013), this study employed item parceling. This approach combines related items from constructs that contain ten or more measurement items (Hagtvet and Nasser 2004). After the collected data underwent basic preprocessing and verification of descriptive statistics, we conducted a confirmatory factor analysis to evaluate the validity of the measurement instrument. Additionally, the study utilized an autoregressive cross-lagged model to examine the longitudinal causal relationships among the variables over time (Ployhart and Vandenberg 2010), using SPSS and AMOS. This method enabled the analysis of how the relationships among managers' BSL, nurses' ACAP, and JE evolved over time (see Figure 3).

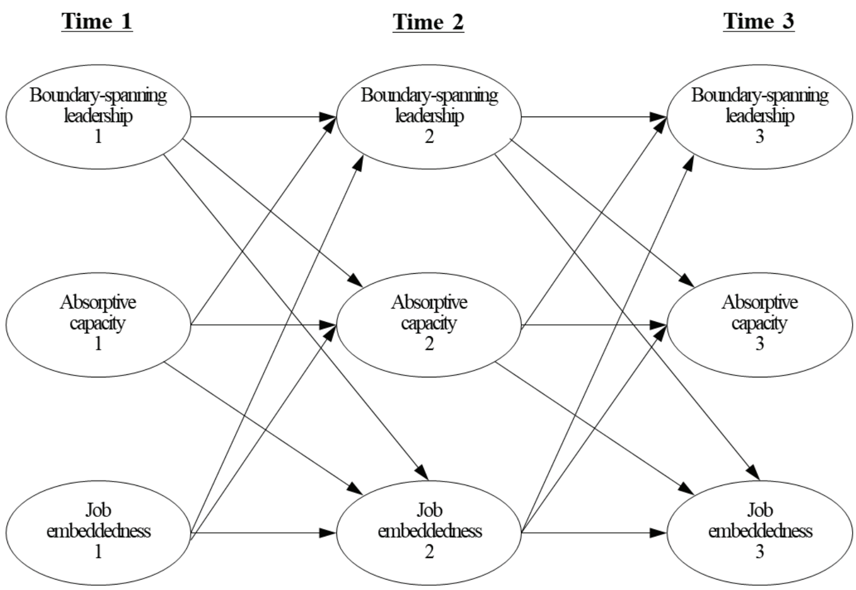

Figure 3: Research Model

Results

The descriptive statistics of BSL, ACAP, and JE across three waves are presented in Table 1. The mean scores for BSL at each respective time point were 4.05 (SD = 0.88), 3.90 (SD = 0.83), and 3.93 (SD = 0.90). According to the results of a one-way ANOVA, the variation in mean values across these time points was not statistically significant for BSL. Similarly, the mean scores of ACAP were 4.17 (SD = 0.93), 4.17 (SD = 0.91), and 4.18 (SD = 0.93) and did not reveal statistically significant differences across the time points based on the ANOVA results. The one-way ANOVA for JE revealed no significant differences in the mean values over time. Regarding JE, the mean scores at each time point were 4.50 (SD = 0.97), 4.43 (SD = 1.02), and 4.55 (SD = 0.93), respectively.

All primary variables exhibited strong correlations among measures within the same time point, and a trend was observed whereby the correlations decreased as the difference in time points increased. For instance, the correlation coefficient between BSL and ACAP at the first time point (T1) was found to be 0.445 ($p<0.001$). The correlation coefficient between BSL at T1 and ACAP at the second time point (T2) was 0.297 ($p<0.001$), and it further dropped to 0.150 ($p<0.05$) between BSL at T1 and ACAP at the third time point (T3). Moreover, the correlation coefficients between BSL and the other two variables were greater than 0.124 ($p<0.05$) across all time points, and those regarding ACAP were at least 0.138 ($p<0.05$). All correlations between JE and the other two variables were significant, except for the correlation between JE at T1 and BSL at T3 (0.095, $p>0.05$). These results suggest that, as argued in prior studies, the three variables maintained strong associations over time, which is consistent with the hypotheses of the present research.

To evaluate the validity of the measurement instrument, we conducted a confirmatory factor analysis to calculate the average variance extracted (AVE) and the composite reliability (CR), as proposed by Farrell (2010) and Bagozzi and Yi (1988). At each of the three time points, the AVE values for all variables exceeded the threshold of 0.5, indicating robust convergent validity (see Table 1). All CR values exceeded the recommended cut-off of 0.6 to 0.7, indicating adequate reliability.

To examine the stability of ACAP, JE, and BSL over time, assess the temporal precedence among these variables, and identify any potential longitudinal mediating effects, we conducted an analysis using a cross-lagged autoregressive model. We commenced by sequentially testing measurement invariance, path invariance, and error covariance invariance to ensure consistent measurements of the same constructs throughout the study period (Wu et al. 2013).

Table 1: Descriptive Statistics and Construct Validity (n = 297)

		T1			*T2*			*T3*			*AVE*	*CR*
		BSL	*ACAP*	*JE*	*BSL*	*ACAP*	*JE*	*BSL*	*ACAP*	*JE*		
T1	BSL	1.000									0.80	0.92
	ACAP	0.445***	1.000								0.83	0.79
	JE	0.327***	0.305***	1.000							0.81	0.73
T2	BSL	0.731***	0.159**	0.129*	1.000						0.68	0.88
	ACAP	0.297***	0.502***	0.184**	0.436***	1.000					0.72	0.83
	JE	0.191***	0.246***	0.522***	0.262***	0.296***	1.000				0.75	0.77
T3	BSL	0.510***	0.144*	0.095	0.624***	0.197***	0.117*	1.000			0.61	0.88
	ACAP	0.150**	0.461***	0.114*	0.328***	0.525***	0.244***	0.425***	1.000		0.61	0.82
	JE	0.124*	0.138*	0.432***	0.186**	0.287***	0.490***	0.236***	0.299***	1.000	0.69	0.82
Mean		4.05	4.17	4.50	3.90	4.17	4.43	3.93	4.18	4.55		
SD		0.88	0.93	0.97	0.83	0.91	1.02	0.90	0.93	0.93		

Notes: * $p<0.05$, ** $p<0.01$, *** $p<0.001$.
Abbreviations: BSL, Boundary-spanning leadership; ACAP, Absorptive capacity; JE, Job embeddedness.

Initially, we used an unconstrained model in which factor loadings were permitted to differ across time points. This unconstrained model provided an adequate fit for all three samples. Subsequently, we imposed constraints to create a constrained model, ensuring that the factor loadings, intercepts, and variances of the residuals were equal over time, in alignment with the concept of strict factorial invariance. We tested $\Delta\chi 2$, ΔRMSEA, and ΔCFI to juxtapose the constrained model with the unconstrained model (Bentler and Satorra 2010). The equality assumption was supported in all models, indicating no significant decline in fit between the constrained and unconstrained models. This supports the consistency of interpretation across the three time points for each sample (Chen 2007; Ployhart and Vandenberg 2010).

The validation of the autoregressive and cross-lagged effects within our proposed model is presented in Table 2 and Figure 4. All autoregressive effects proved statistically significant, indicating that the preceding levels of BSL, ACAP, and JE effectively predicted the same variables at subsequent time points. Specifically, BSL at T1 significantly predicted BSL at T2 ($\beta = 0.852$, $p<0.001$), and BSL at T2 also significantly predicted BSL at T3 ($\beta = 0.552$, $p<0.001$). These results substantiate the persistence of nurses' perceptions of managerial BSL throughout the study period. Next, ACAP sustained its significant influences, from T1 to T2 ($\beta = 0.435$, $p<0.001$) and from T2 to T3 ($\beta = 0.513$, $p<0.001$). Finally, JE consistently maintained its significant effects, from T1 to T2 ($\beta = 0.484$, $p<0.001$) and from T2 to T3 ($\beta = 0.439$, $p<0.001$).

Table 2: Autoregressive effect and cross-lagged effect (n = 297)

Effects	*Regressors*	*Path*	*B*	*β*	*S.E.*	*Z*
Autoregressive effect	BSL	BSL(T1) → BSL(T2)	0.804	0.852***	0.073	11.045
		BSL(T2) → BSL(T3)	0.597	0.552***	0.074	8.037
	ACAP	ACAP(T1) → ACAP(T2)	0.426	0.435***	0.053	8.025
		ACAP(T2) → ACAP(T3)	0.525	0.513***	0.054	9.739
	JE	JE(T1) → JE(T2)	0.509	0.484***	0.048	10.663
		JE(T2) → JE(T3)	0.403	0.439***	0.045	9.026
Cross-lagged effect	BSL	BSL(T1) → ACAP(T2)	0.292	0.283***	0.080	3.626
		BSL(T1) → JE(T2)	0.140	0.121	0.077	1.821
		BSL(T2) → ACAP(T3)	0.377	0.337***	0.080	4.732
		BSL(T2) → JE(T3)	0.144	0.129	0.078	1.841
	ACAP	ACAP(T1) → BSL(T2)	0.098	0.110*	0.047	2.077
		ACAP(T1) → JE(T2)	0.123	0.113*	0.054	2.273
		ACAP(T2) → BSL(T3)	0.095	0.102*	0.048	1.982
		ACAP(T2) → JE(T3)	0.242	0.237**	0.054	4.516
	JE	JE(T1) → BSL(T2)	0.038	0.044	0.040	0.932
		JE(T1) → ACAP(T2)	0.009	0.009	0.047	0.188
		JE(T2) → BSL(T3)	0.010	0.011	0.040	0.252
		JE(T2) → ACAP(T3)	0.046	0.050	0.047	0.976

Notes: * $p<0.05$, ** $p<0.01$, *** $p<0.001$.
Abbreviations: BSL, boundary-spanning leadership; ACAP, Absorptive capacity; JE, Job embeddedness.

The results regarding the cross-lagged effects between BSL and ACAP are as follows: BSL at each time point significantly and positively predicted ACAP at the subsequent time points. Specifically, BSL at T1 significantly predicted ACAP at T2 ($\beta = 0.283$, $p<0.001$), and BSL at T2 significantly predicted ACAP at T3 ($\beta = 0.337$, $p<0.001$). Conversely, ACAP at each time point significantly predicted BSL at subsequent time points. ACAP at T1 significantly predicted BSL at T2 ($\beta = 0.110$, $p<0.05$), and ACAP at T2 significantly predicted BSL at T3 ($\beta = 0.102$, $p<0.05$). Contrary to our initial hypotheses, this evidence corroborated the reciprocal relationship between BSL and ACAP.

Next, the results for cross-lagged effects between ACAP and JE are as follows: ACAP at T1 positively predicted JE at T2 ($\beta = 0.113$, $p<0.05$), and ACAP at T2 positively predicted JE at T3 ($\beta = 0.237$, $p<0.01$). Conversely, JE at T1 did not significantly predict ACAP at T2 ($\beta = 0.009$, $p>0.05$), and JE at T2 likewise did not significantly predict ACAP at T3 ($\beta = 0.050$, $p>0.05$).

The cross-lagged effects between BSL and JE were not significant. Specifically, the effects of BSL at T1 on JE at T2 ($\beta = 0.121$, $p>0.05$) and BSL at T2 on JE at T3 ($\beta = 0.129$, $p>0.05$) were both non-significant. Similarly, the effects of JE at T1 on BSL at T2 ($\beta = 0.044$, $p>0.05$) and JE at T2 on BSL at T3 ($\beta = 0.011$, $p>0.05$) were both non-significant.

To examine the longitudinal mediating effect of ACAP on the relationship between managers' BSL and nurses' JE, we employed a bootstrapping approach (Shrout and Bolger 2002). The mediating influence of ACAP at T2 between BSL at T1 and JE at T3 was found to be non-significant, as indicated by the 95% confidence interval ranging from -0.237 to 0.382, which included zero.

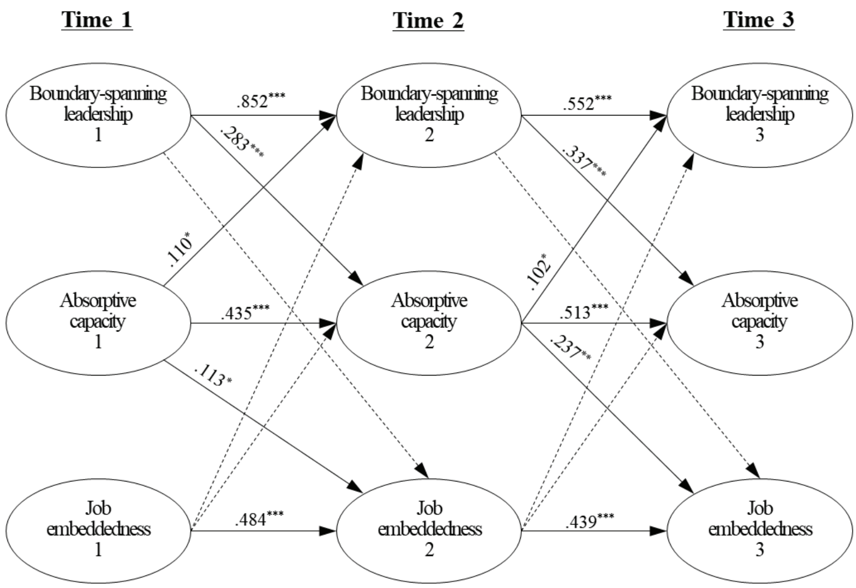

Figure 4. Results of Autoregressive Cross-Lagged Model Analysis

Notes. * $p<0.05$, ** $p<0.01$, *** $p<0.001$. For simplicity, only the standardized coefficients are presented.

Discussion

This research was initiated by recognizing the need to explore leadership styles that could improve the deteriorating JE of nurses during the COVID-19 pandemic. In this context, we proposed a hypothetical research model suggesting that BSL fosters and activates nurses' ACAP, thereby enhancing JE. Previous studies examining the relationships between BSL, ACAP, and JE are scarce. Even the few existing studies related to each variable have primarily relied on cross-sectional data, which limits the examination of changes over time and the causal relationships among these three variables. To address this limitation, we employed an autoregressive cross-lagged model to verify the stability of these variables over time and to test the causal directionality among them. The major findings and implications of this study are elaborated as follows.

First, the results from the analysis of the autoregressive effect revealed that the constructs of BSL, ACAP, and JE remained stable over time. Specifically, the levels of BSL, ACAP, and JE at a preceding time point were found to predict their respective levels at a subsequent time point with statistical significance. This indicates that these constructs are not merely temporary perceptions but consistently persist over time. Notably, BSL at the managerial level and ACAP at the individual level are relatively recent concepts in business management and have been subject to varying interpretations among scholars. However, our findings are academically significant, as they confirm that these three variables possess discriminant validity, demonstrating distinctiveness and consistent perceptions over time.

Second, the longitudinal cross-lagged effect of BSL on JE was not statistically significant, contrary to our hypothesis. This suggests that the impact of prior BSL on JE at a subsequent time point was negligible. However, the cross-lagged effects between BSL and ACAP, as well as between ACAP and JE, were found to be statistically significant. Consequently, while the direct effect of BSL on JE was not significant, its indirect effect, mediated via ACAP, is deemed meaningful. The complete mediating role of ACAP between BSL and JE implies that enhancing JE can be enhanced when nurses are provided with opportunities and experiences to enact their ACAP through effective managerial BSL.

Third, the bidirectional cross-lagged effects between BSL and ACAP suggest that these two variables reciprocally influence each other over time. Evidence supporting our research hypothesis emerged, showing that a higher level of BSL at a preceding time point was positively related to an elevated level of ACAP at a subsequent time point. BSL, by encouraging behaviors that activate both internal and external organizational networks, appears to positively influence the development of nurses' ACAP. Interestingly, our results revealed an unexpected pattern: a higher level of ACAP at a previous time point led to an increased level of BSL at the subsequent time point. This indicates that as nurses' ACAP increases, the potential for managers to exhibit effective BSL also grows.

This research achieved theoretical advancement by finding new meaning in this bidirectional influence that arises from the synchronization of the maturity stages of BSL and ACAP. Both BSL and ACAP are characterized by distinct stages throughout their implementation (Yildiz et al. 2019). ACAP begins at the recognition stage, where external knowledge is identified and captured. It then progresses to the assimilation stage, during which the acquired knowledge is analyzed, processed, interpreted, and integrated with existing knowledge. It culminates at the exploitation stage, where assimilated knowledge is used to enhance existing capabilities. Although recognition and assimilation stages may not directly lead to outcomes, these stages provide the foundation for exploitation. Similarly, BSL unfolds through three strategies: managing boundaries, forging common ground, and discovering new frontiers (Ernst and Chrobot-Mason 2010). The strategy of managing boundaries involves reflecting and buffering, while forging common ground encompasses mobilizing and connecting. Discovering new frontiers incorporates transforming and weaving aspects.

Given the multi-stage characteristics of both ACAP and BSL, it is plausible that the quality and level of activities pursued by a manager's BSL closely align with the quality and level of ACAP activities carried out by nurses. If nurses' ACAP remains at the assimilation stage, it may be difficult for managers to execute BSL at its most advanced strategy—'discovering new frontiers.' Conversely, this relationship is expected to hold true reciprocally. Consequently, BSL and ACAP can be inferred to have a positive, mutually influential relationship that depends on the level at which each is enacted.

Based on the findings of this study, several practical implications can be drawn. First, this research offers practical insights for enhancing BSL and ACAP in the corporate environment by leveraging the bidirectional influence that arises from their synchronized maturity stages. It is necessary for hospital administrators to emphasize the role of BSL in regular feedback sessions and workshops that promote knowledge sharing and the exploration of new ideas, as well as in tailored training programs. Managers with BSL can foster communication through boundary management strategies to promote a collaborative atmosphere, while these interventions help employees effectively integrate external knowledge. Additionally, implementing a system to monitor the interactions between ACAP and BSL can ensure that the development of ACAP positively facilitates the execution of BSL. These initiatives collectively enhance the organization's innovation capacity and support long-term performance.

Second, the empirical evidence suggesting that a manager's BSL in hospital can enhance JE among nurses through ACAP, highlights a new role for BSL. BSL is fundamentally linked to the development of networks that facilitate the acquisition of information and knowledge via flexible boundary management. Although this approach may increase workload by requiring nurses to engage in unspecified extra-role behaviors, it has been demonstrated that ACAP positively influences JE. This indicates a pressing need for hospital administrators and policymakers to establish a leadership role focused on boundary management to improve JE among nurses, thereby underscoring the importance of systematically fostering nurses' ACAP.

Third, this study offers empirical evidence that elucidates the previously theoretical understanding of the influential relationship between BSL and ACAP by revealing their bidirectional dynamic. A practical framework for managing BSL and ACAP in a reciprocal fashion is essential. To maintain the mutually reinforcing relationship between these variables, it is essential to develop measurement tools or consulting methodologies that can evaluate the qualitative level of employees' ACAP. This highlights the need for hospital administrators, HR leaders, and policymakers to continuously evaluate leadership development programs aimed at nurturing and strengthening effective BSL.

The limitations of this study, along with suggestions for future research, are as follows. First, the generalizability of the findings is limited by the exclusive focus on nurses in eleven Chinese hospitals. Nurses were selected due to the relative ease of measuring JE in a profession known for high turnover rates. Future research would benefit from a more rigorous design that extends the application to a broader range of occupations.

Second, this study showed a sample retention rate of 68.3% from the initial to the final wave, indicating a relatively high dropout rate. Although some attrition is inevitable in longitudinal studies, the relatively low retention rate may limit the representativeness of the final sample. Future studies should develop strategies to maintain high sample retention rates and consider examining the characteristics of participants who drop out prematurely.

Lastly, although this study used longitudinal data to overcome the constraints of cross-sectional studies, it is still limited by the relatively short measurement interval compared to other longitudinal studies. The three-month interval was established to capture short-term changes in the research variables—BSL, ACAP, and JE—within routine hospital interactions. However, the short measurement interval may have introduced learning effects in participants' responses, and the causal directionality and mediating effects among variables may not have been distinctly delineated. Thus, future studies should consider longer study durations and extended measurement intervals to examine the relationships between each variable from a broader temporal perspective.

Conclusion

During the COVID-19 pandemic, hospitals faced a critical need for boundary-spanning behavior, which, combined with increased workloads, contributed to higher turnover rates. Therefore, this study investigated the influence of a manager's BSL on nurses' ACAP, particularly in terms of the acquisition, dissemination, and exploitation of knowledge and information both internally and externally within the organization. Additionally, it explored the subsequent impact of this relationship on enhancing nurses' JE. Although prior research on the relationships among BSL, ACAP, and JE has been limited, this study rigorously assessed their temporal stability and causal relationships using an autoregressive cross-lagged model. First, the autoregressive effects confirmed the temporal stability of BSL, ACAP, and JE constructs. Second, the analysis of the cross-lagged effects supported the hypothesis that improvements in nurses' ACAP, facilitated by a manager's BSL, could lead to enhanced JE. Third, the identification of a bidirectional relationship between BSL and ACAP revealed a previously unrecognized reciprocal dynamic between these two constructs. Based on these findings, practical implications can be drawn.

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Informed Consent

The authors have obtained informed consent from all participants.

Conflict of Interest

The authors declare that there is no conflict of interest.

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